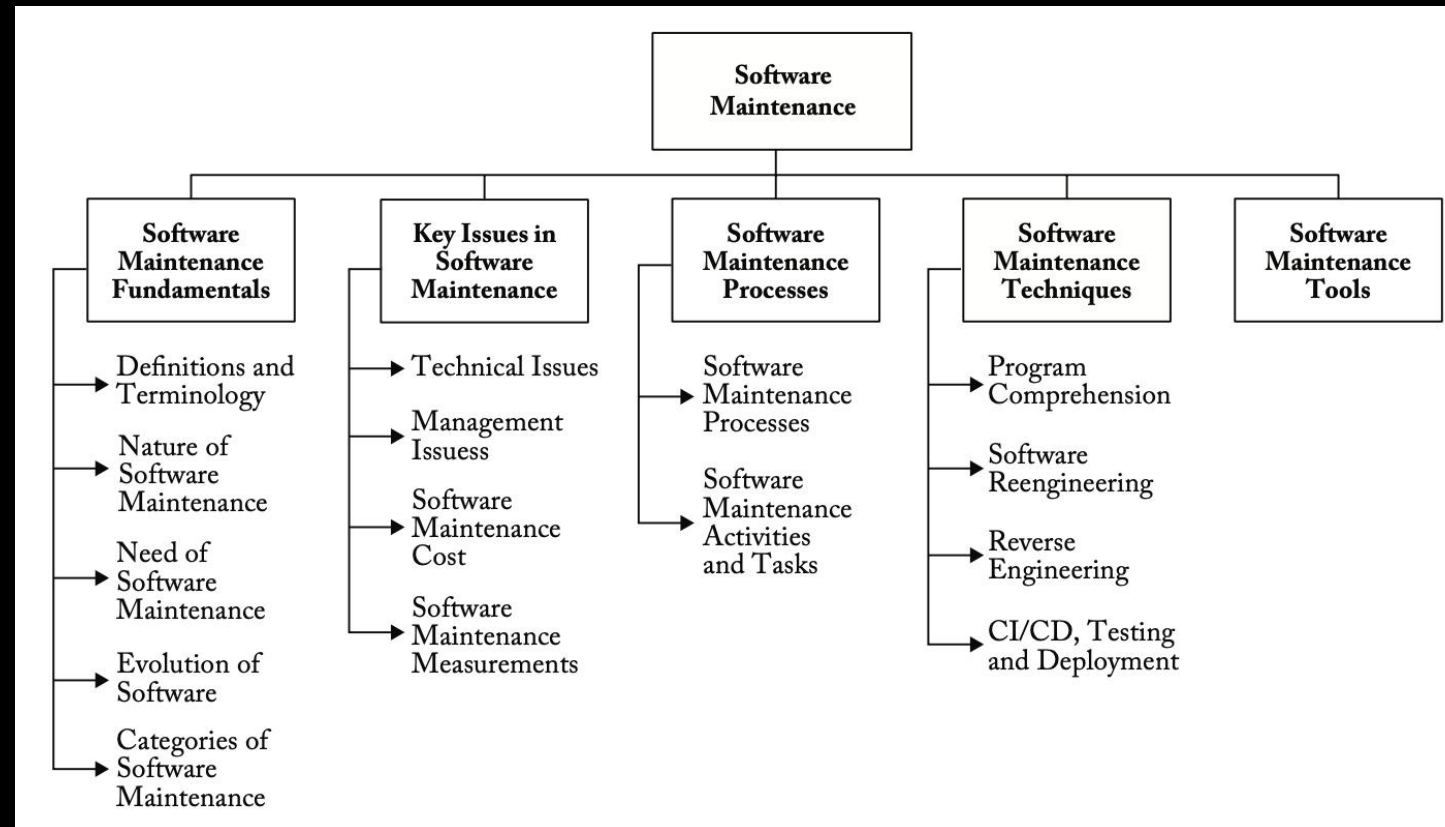


Software Maintenance & Quality Assurance

TTTE6104

Software Engineering
Paradigm

Breakdown of Topic



ACRONYMS

API	Application Programming Interface
CI	Continuous Integration
IEC	The International Electrotechnical Commission
IEEE	The Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
KA	Knowledge Area
LOC	Lines of Code
MR	Modification Request
PR	Problem Report
SCM	Software Configuration Management
SEE	Software Engineering Environment
SLA	Service-Level Agreement
SLI	Service-Level Indicators
SLO	Service-Level Objectives
SQA	Software Quality Assurance
V&V	Verification and Validation
XaaS	Anything as a Service

What is software maintenance?

- Maintenance is a technical process in the life cycle of a software product.
- It contains the activities and tasks of the maintenance organization.
- During operation of the software, problems may be detected that were not detected during verification, validation and acceptance. Therefore, a maintenance effort is needed to cope with these problems.
- Needed when upgrading system components.
- The maintenance phase usually lasts until the software is retired by the disposal process

Need for Software Maintenance

- Correct faults and latent defects
- Improve the design or performance of operational software
- Implement enhancements
- Help users understand the software's functionality
- Adapt to changes in interfaced systems or infrastructure
- Prevent security threats
- Remediate technical obsolescence of system or software elements
- Retire the software

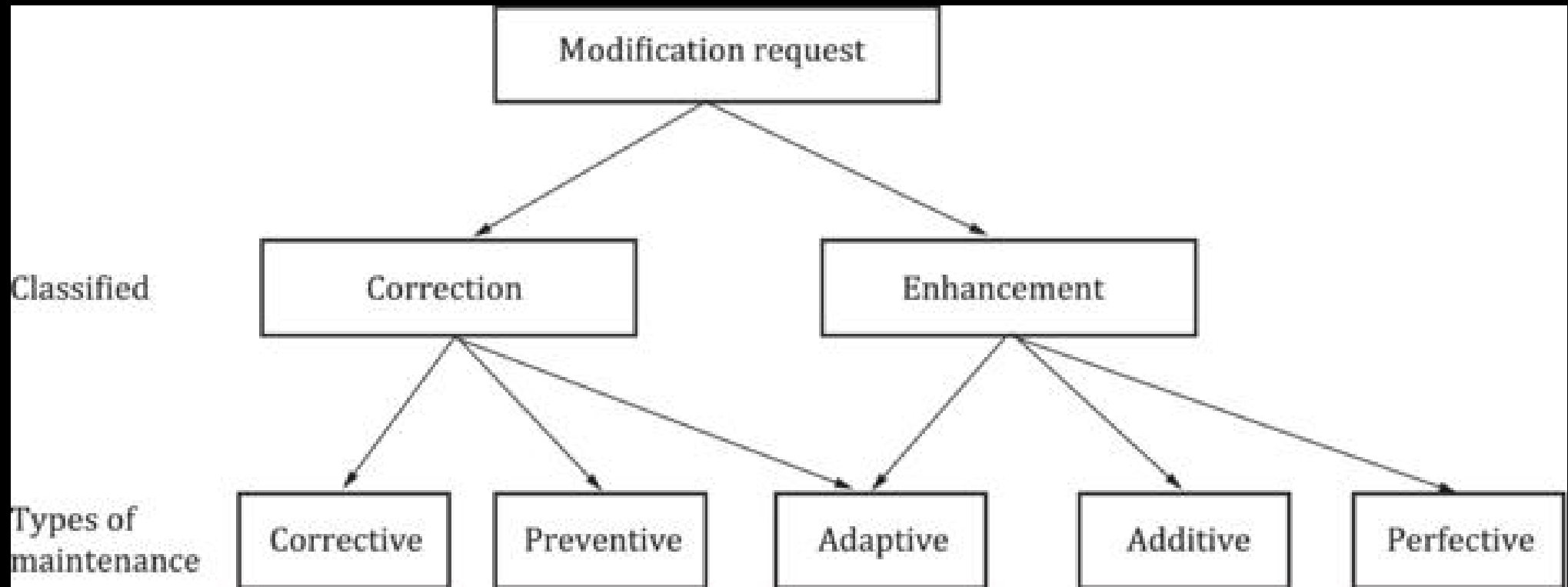
Maintenance Costs

- Operating environment (hardware and software).
- Organizational environment (policies, competition, process, product and personnel).

Evolution of Software

- Continuing Change
- Increasing Complexity
- Self-Regulation
- Invariant Work Rate
- Conservation of Familiarity
- Continuing Growth
- Declining Quality
- Feedback System

Modification Request



Acitvity

- **Instruction:**

Choose **one real software system** that you personally use every day (for example: WhatsApp, Shopee, Maybank2u, Grab, Facebook, Instagram, Google Maps).

- Search online / observe app update notes / reflect from your experience and identify at least **three** recent changes or updates that happened to that system.
- For each change, classify the type of maintenance:

Change / Update	Type of Maintenance	Why you classify it as that type?
e.g., fixing login error	Corrective maintenance	System failed to login → bug fix

- **Deliverable:**

Create a simple table summarising your findings.

Categories of Software Maintenance

- Five categories (types) of software maintenance have been standardized to classify a maintenance request: corrective, preventive, adaptive, additive and perfective.
- ISO/IEC/IEEE 14764 [1] also defines a sixth category — emergency maintenance.

Key Issues in Software Maintenance

- Technical issues
 - Limited understanding, testing, impact analysis, maintainability
- Management issues
 - Alignment with Organizational Objectives, staffing, process, supplier management, Organizational Aspects of Maintenance
- Software maintenance costs
 - Technical Debt Cost Estimation, Maintenance Cost Estimation,
- Software maintenance measurement

Software Maintenance Processes

- Prepare for maintenance
- Perform maintenance
- Perform logistics support
- Manage results of maintenance and logistics

Processes, activities and practices that are specialized to software maintenance

- Program understanding
- Transition
- MR acceptance/rejection
- Maintenance help desk
- Impact analysis
- Maintenance service-level indicators (SLIs), service-level objectives (SLOs), SLAs, and maintenance software and hardware licenses and contracts

Planning

- Business planning (organizational level)
- Maintenance planning (transition level)
- Release/version planning (software level)
- MR planning (at individual request level).
 - Collect the dates of availability of individual requests.
 - Agree with users on the content of sub–sequent releases/versions.
 - Identify potential conflicts and develop alternatives.
 - Assess the risk of a given release and develop a back–out plan in case problems arise
 - Inform all stakeholders.

Software Maintenance Techniques

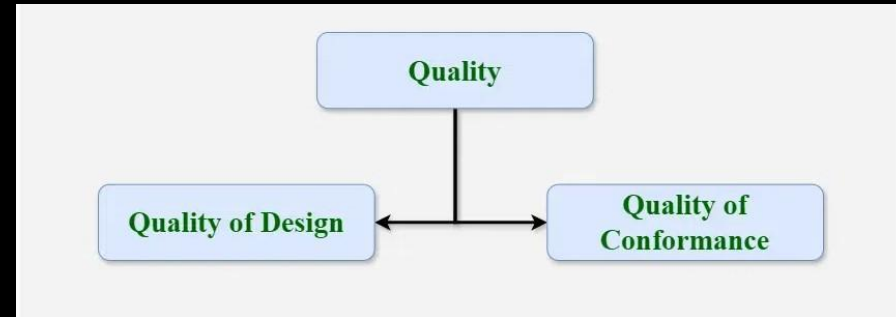
- Program Comprehension
- Software Reengineering
- Reverse Engineering
- Continuous Integration, Delivery, Testing and Deployment
- Visualizing Maintenance

Software Maintenance Tools

- Configuration management, code versioning and code review tools
- software testing tools
- software quality assessment tools
- Program slicers
- Static analyzers
- Dynamic analyzers
- Data flow analyzers
- Cross-referencers
- Dependency analyzers
- Remote Access tools

Software Quality Assurance

- A set of activities that ensure processes, procedures as well as standards are suitable for the project and implemented correctly.
- It focuses on improving the process of development of software – problems can be prevented before they become major issues.
- The quality of the software is verified by third-party organizations – ISO Standards in Software Engineering



Elements of Software Quality Assurance

- Standards
- Reviews and audits
- Testing
- Error/defect collection and analysis
- Change management
- Education
- Security management
- Safety
- Risk management

Focus of Software Quality Assurance

Software Quality Assurance (SQA) focuses



Major Activities in Software Quality Assurance

- SQA Management Plan
 - Set The Check Points
 - Measure Change Impact
 - Multi testing Strategy
 - Manage Good Relations
 - Maintaining records and reports
-
- Reviews software engineering activities
 - Formalize deviation handling

References

- Sommerville, I. (2016). Software engineering. Pearson.
- <https://www.iso.org/obp/ui/en/#iso:std:iso-iec-ieee:14764:ed-3:v1:en>
- Guide to the Software Engineering Body of Knowledge
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